



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 01

- 1. Title:** Demonstrate inline, internal, and external JavaScript, use console methods; create a webpage showing user info and a welcome message.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), Web Browser (Google Chrome / Microsoft Edge), HTML5, JavaScript, Windows Operating System.
- 3. Theory:** JavaScript is a client-side scripting language used to make web pages interactive and dynamic. It can be added to a webpage in three ways:
 - **Inline JavaScript:** Code written directly inside HTML elements using event attributes.
 - **Internal JavaScript:** Code written inside the `<script>` tag within the HTML file.
 - **External JavaScript:** Code written in a separate .js file and linked to the HTML document.
 - In this experiment, JavaScript is used to display a welcome message, show user information, and print output using console methods such as `console.log ()`, `console. warn()`, and `console.error()`.



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4. Output:

File C:/Users/vithi/OneDrive/Desktop/java3.html

Vithika Saoji
B.Tech CSE (Data Science) Student

About Skills Education Projects Contact

About Me

Hello! My name is Vithika Saoji. I am a second-year B.Tech Computer Science and Engineering student specializing in Data Science at Symbiosis Institute of Technology, Nagpur. I enjoy learning new technologies, web development, cloud computing, cybersecurity, and programming. I love building projects and improving my technical skills.

My Skills

- C Programming
- Python
- Java
- HTML
- CSS
- SQL
- AWS Basics
- Problem Solving
- Teamwork

Education

College: Symbiosis Institute of Technology, Nagpur
Degree: B.Tech Computer Science & Engineering
Specialization: Data Science

Projects

- AutoVAPT Platform
- Disease Prediction Using Machine Learning
- Air Quality Monitoring System

Contact Me

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Phone: +91 XXXXXXXXXX

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5. Result/Conclusion: The webpage was successfully created using HTML and JavaScript. It demonstrated the use of inline, internal, and external JavaScript. User information and a welcome message were displayed correctly, and console methods produced the expected output in the browser's developer console. This experiment helped understand different ways of implementing JavaScript and using console methods for debugging.



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Case Study No.:

1. Case Study Title: Create a webpage for college department and link the student information page with it and write a java script to display “welcome to sit nagpur” in an alert box.

2. Case Study Program Code:

```
File Edit Selection View Go Run Terminal Help
JS script.js C:\...Desktop index1.html launch.json main.html index.html JS script.js JAVA.html student_info.html
index.html > ...
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>SIT Nagpur - Department of Computer Science</title>
7   <style>
8     /* Internal CSS */
9     body { font-family: Arial, sans-serif; margin: 0; padding: 0; background-color: #f4f7f6; }
10    header { background-color: #004085; color: white; padding: 20px; text-align: center; }
11    nav { background-color: #005cbf; padding: 10px; text-align: center; }
12    nav a { color: white; margin: 0 15px; text-decoration: none; font-weight: bold; }
13    nav a:hover { text-decoration: underline; }
14    .container { padding: 20px; text-align: center; }
15    button { padding: 10px 20px; font-size: 16px; margin: 10px; cursor: pointer; background-color: #28a745; color: white; border: none; border-radius: 5px; }
16    button:hover { background-color: #218838; }
17  </style>
18 </head>
19 <body>
20
21   <header>
22     <h1>Welcome to Department of IT</h1>
23     <h2>SIT NAGPUR</h2>
24   </header>
25
26   <nav>
27     <a href="index.html">Home</a>
28     <a href="student_info.html">Student Information</a>
29   </nav>
30
31   <div class="container">
32     <h3>About Our Department</h3>
33     <p>Providing cutting-edge technology education and empowering future innovators at SIT Nagpur.</p>
34   </div>
35 </body>
36 </html>
```



```
File Edit Selection View Go Run Terminal Help

JS script.js C:\...\Desktop index1.html launch.json main.html index.html

JAVA.html > JS script.js > ...
1 // This function will be triggered by the inline button
2 function displayInlineAlert() {
3     alert("Welcome to SIT NAGPUR");
4 }
5
6 // Console logging for debugging using Browser Developer Tools
7 console.log("External JavaScript loaded successfully.");
8 console.log("Welcome to SIT NAGPUR - logged from the console!");
9
```

```
File Edit Selection View Go Run Terminal Help qwertyuiop

JS script.js C:\...\Desktop index1.html launch.json main.html index.html JS script.js JAVA.html student_info.html

JAVA.html > student_info.html > ...
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <meta charset="UTF-8">
5     <title>Student Information - SIT NAGPUR</title>
6     <style>
7         body { font-family: Arial, sans-serif; padding: 20px; background-color: #f9f9f9; }
8         .container { max-width: 600px; margin: 0 auto; background: white; padding: 20px; box-shadow: 0 0 10px rgba(0,0,0,0.1); }
9         a { display: block; margin-top: 20px; text-decoration: none; color: #005cbf; }
10    </style>
11 </head>
12 <body>
13
14     <div class="container">
15         <h2>Student Information Portal</h2>
16         <p>Access student records, timetables, and academic performance data here.</p>
17
18         <form>
19             <label for="roll">Enter Student Roll No:</label><br><br>
20             <input type="text" id="roll" name="roll" required>
21             <button type="button" onclick="searchStudent()">Search Record</button>
22         </form>
23
24         <a href="index.html">Back to Home</a>
25     </div>
26
27     <script>
28         // Internal JavaScript specifically for this page
29         function searchStudent() {
30             let rollNo = document.getElementById('roll').value;
31             console.log("Searching database for Roll No: " + rollNo);
32             alert("Record search initiated for Roll No: " + rollNo);
33         }
34     </script>
35
```



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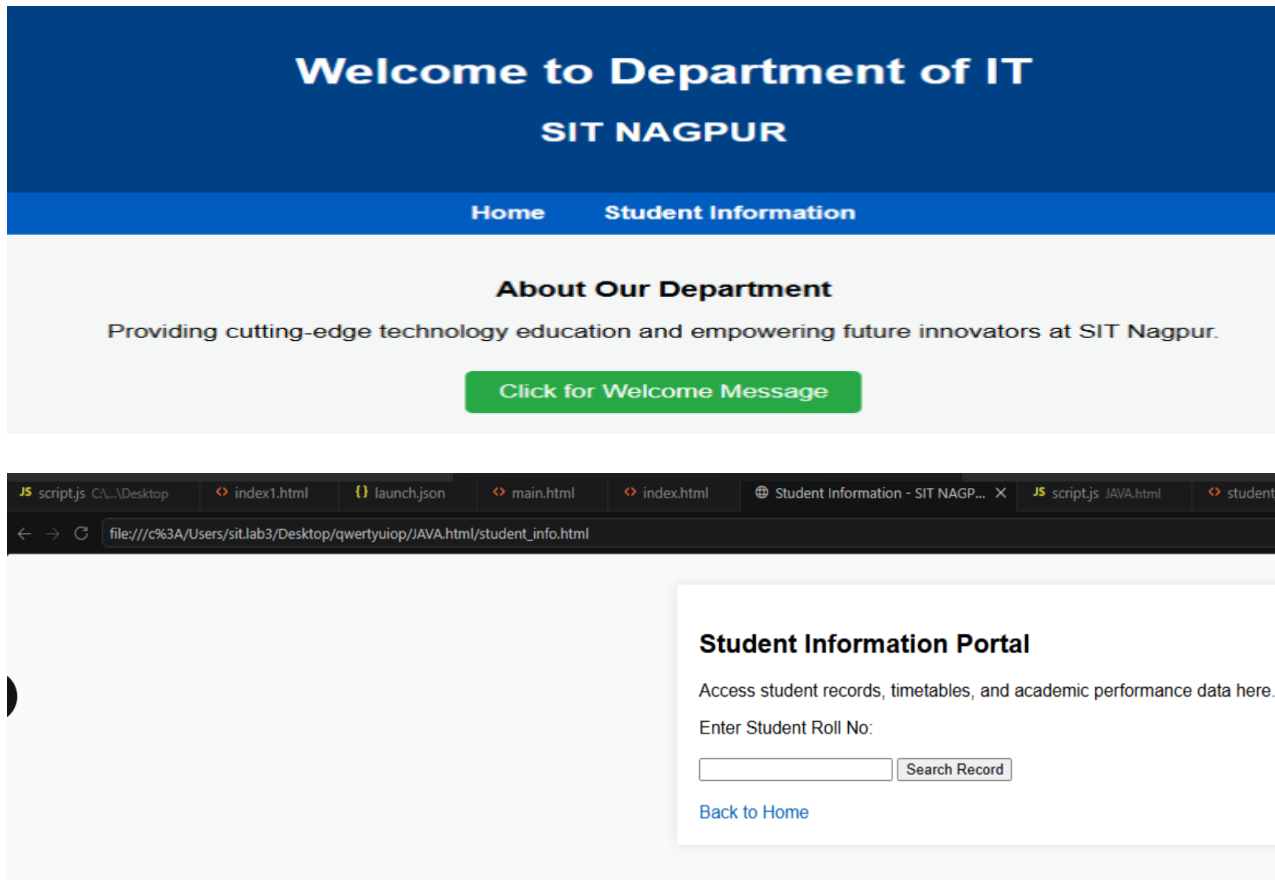
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3. Output:



4. Result/Conclusion: The college department webpage was created successfully. The Student Information page was linked to it, and JavaScript was implemented using inline, internal, and external methods. A welcome message was displayed in an alert box, and `console.log()` output was verified in the browser console.



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Experiment No.: 2

1. Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: `var` is function-scoped, `let` is block-scoped, and `const` is used for fixed values.

The `prompt()` function accepts user input, and `Number()` converts input into numeric values.

The program calculates the total amount, GST, and final bill using arithmetic operations.

An object stores all billing details together, while object destructuring extracts required values easily.

Template literals (``) are used to display formatted output with variable values using `$${}`.

The `document.write()` method displays the billing receipt on the webpage.

4. Output :

← × ⓘ File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/index.html

This page says

Enter Customer Name:

OK

Cancel

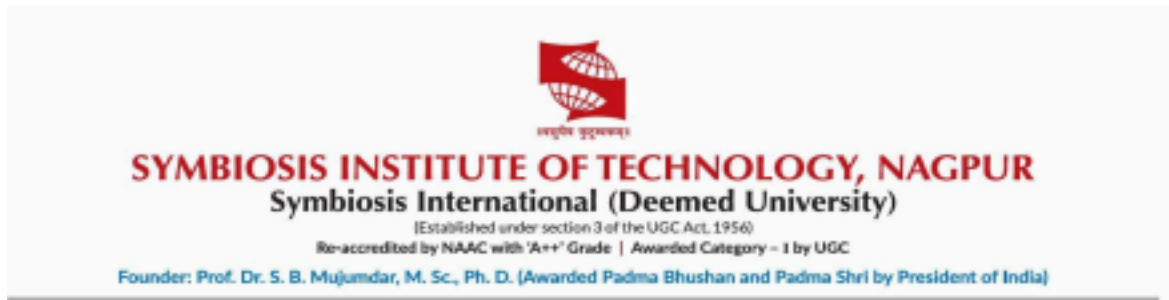
← × ⓘ File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/index.html

This page says

Enter Product Name:

OK

Cancel



File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/index.html

This page says

Enter Quantity:

OK

Cancel

File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/index.html

// Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Billing Calculator

Billing Receipt

Customer: Vithika

product: Laptop

Quantity: 3

Price per Item: ₹70000

GST (18%): ₹37800.00

Final Bill: ₹247800.00

5. Result/Conclusion: A billing calculator was successfully created using JavaScript. The program accepted user input, calculated the bill with GST, and demonstrated the use of var, let, const, template literals, and object destructuring effectively.



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Case Study No.:

1. Case Study Title: Develop the webpage for an online shopping website using HTML CSS and HTML5 tags.

2. Case Study Program Code:

```
Welcome JS Practical2.js index.html
Case study 2 java.html > html > head > style > .container
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Online Shopping Bill</title>
7
8 <style>
9 body{
10     font-family:Arial,sans-serif;
11     background: #f2f2f2;
12 }
13 .container{
14     width:500px;
15     margin:30px auto;
16     background: white;
17     padding:20px;
18     border-radius:10px;
19     box-shadow:0 0 10px gray;
20 }
```

```
20 }
21 h2{
22     text-align:center;
23     color: darkblue;
24 }
25 label{
26     display:block;
27     margin-top:10px;
28     font-weight:bold;
29 }
30 input,select{
31     width:100%;
32     padding:8px;
33     margin-top:5px;
34 }
35 button{
36     width:100%;
37     margin-top:20px;
38     padding:10px;
39     background: green;
40     color: white;
41     border:none;
42     font-size:18px;
43     cursor:pointer;
44 }
45 button:hover{
46     background: darkgreen;
47 }
48 #result{
```

```

Case study 2.java.html > html > head > style
2  <html lang="en">
3  <head>
8  <style>
48  #result{
49      margin-top:20px;
50      padding:15px;
51      background: #eef;
52  }
53  </style>
54  </head>
55  <body>
56
57  <div class="container">
58
59  <h2>Online Shopping Bill</h2>
60
61  <label>Customer Name</label>
62  <input type="text" id="name">
63
64  <label>Mobile Number</label>
65  <input type="text" id="mobile">
66
67  <label>Invoice Number</label>
68  <input type="text" id="invoice">
69
70  <label>Product Name</label>
71  <input type="text" id="product">
72
73  <label>Quantity (Kg)</label>
74  <input type="number" id="qty">
75
76  <label>Rate per Kg (₹)</label>
77  <input type="number" id="rate">
78

```

```

Case study 2.java.html > html > body > div.container > button
2  <html lang="en">
55  <body>
57  <div class="container">
79  <label>Discount (%)</label>
80  <input type="number" id="discount">
81
82  <label>GST (%)</label>
83  <input type="number" id="gst">
84
85  <label>Packing Charges (₹)</label>
86  <input type="number" id="packing">
87
88  <label>Payment Mode</label>
89  <select id="payment">
90  <option>Cash</option>
91  <option>Card</option>
92  <option>UPI</option>
93  </select>
94
95  <label>Membership</label>
96  <select id="membership">
97  <option>Yes</option>
98  <option>No</option>
99  </select>
100
101  <button onclick="calculateBill()">Calculate Bill</button>
102  <div id="result"></div>
103  </div>
104  <script>
105
106  var shopName = "Fresh Mart";
107  let gstApplicable = true;
108  const country = "India";

```

```

Case study 2 java.html > html > body > script > calculateBill
2   <html lang="en">
55  <body>
104 <script>

109 let demoUndefined;
110 let demoNull = null;
111
112 function calculateBill(){
113
114     let name=document.getElementById("name").value;
115     let mobile=document.getElementById("mobile").value;
116     let invoice=document.getElementById("invoice").value;
117     let product=document.getElementById("product").value;
118     let qty=Number(document.getElementById("qty").value);
119     let rate=Number(document.getElementById("rate").value);
120     let discount=Number(document.getElementById("discount").value);
121     let gst=Number(document.getElementById("gst").value);
122     let packing=Number(document.getElementById("packing").value);
123     let payment=document.getElementById("payment").value;
124     let membership=document.getElementById("membership").value;
125     if(membership=="Yes"){
126         discount+=5;
127     }
128     let amount=qty*rate;
129     let discountAmount=(amount*discount)/100;
130     let afterDiscount=amount-discountAmount;
131     let gstAmount=(afterDiscount*gst)/100;
132     let finalBill=afterDiscount+gstAmount+packing;
133
134     document.getElementById("result").innerHTML=`
135
136     <h3>Invoice</h3>
137     <p><b>Shop Name:</b> ${shopName}</p>
138     <p><b>Customer Name:</b> ${name}</p>
139     <p><b>Mobile Number:</b> ${mobile}</p>
140     <p><b>Invoice Number:</b> ${invoice}</p>
141     <p><b>Product Name:</b> ${product}</p>
142     <p><b>Quantity:</b> ${qty} Kg</p>
143     <p><b>Rate:</b> ₹${rate}</p>
144     <p><b>Amount:</b> ₹${amount.toFixed(2)}</p>
145     <p><b>Discount:</b> ₹${discountAmount.toFixed(2)}</p>
146     <p><b>GST:</b> ₹${gstAmount.toFixed(2)}</p>

```

```

147     <p><b>Packing Charges:</b> ₹${packing.toFixed(2)}</p>
148     <p><b>Payment Mode:</b> ${payment}</p>
149     <p><b>Membership:</b> ${membership}</p>
150     <hr>
151
152     <h2>Total Bill : ₹${finalBill.toFixed(2)}</h2>
153     `;
154
155     console.log("Bill Generated");
156     console.info("Customer : "+name);
157     console.warn("GST Applied");
158     console.error("Demo Error Message");
159
160     console.log("Type Conversion:", Number("100"));
161     console.log("Type Coercion:", "100"+20);

```



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3. Output:

File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/Case%20study%20%20ja

Online Shopping Bill	
Customer Name	Vithika Saoji
Mobile Number	9756463728
Invoice Number	7657483928
Product Name	headphone
Quantity (Kg)	7
Rate per Kg (₹)	3500
Discount (%)	150
GST (%)	2
Packing Charges (₹)	50
Payment Mode	Cash
Membership	Yes
Calculate Bill	

File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/Case%20study%20%20java.html

Invoice
Shop Name: Fresh Mart
Customer Name: Vithika Saoji
Mobile Number: 9756463728
Invoice Number: 7657483928
Product Name: headphone
Quantity: 7 Kg
Rate: ₹3500
Amount: ₹24500.00
Discount: ₹37975.00
GST: ₹-269.50
Packing Charges: ₹50.00
Payment Mode: Cash
Membership: Yes
Total Bill : ₹-13694.50



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4. Result/Conclusion: The online shopping billing webpage was successfully developed using HTML, CSS, and JavaScript. It accurately accepts customer and product details, calculates the final bill by applying discount, GST, and packing charges, and demonstrates the use of JavaScript data types, variable declarations, type conversion, and type coercion.



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Experiment No.: 03

1. Title: Implement control structures and form validation; create a grading system based on user-entered marks.

2. Software/Tools Required: Visual Studio Code (VS Code), Web Browser (Google Chrome / Microsoft Edge), HTML, JavaScript, Windows Operating System.

3. Theory:

This program demonstrates the use of control structures and form validation in JavaScript. An HTML form is used to accept marks from the user. Validation checks whether the input is empty and ensures the entered marks are between 0 and 100. If the input is invalid, an alert message is displayed. If the input is valid, if-else control statements are used to determine the grade (A, B, C, D, or F) based on the entered marks, and the result is displayed on the webpage.

4. OUTPUT:



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File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/Practical%203%20java%20

5. Result/Conclusion: The grading system was successfully implemented using HTML, CSS, and JavaScript. The program validates user input, prevents invalid entries, and accurately assigns grades based on the entered marks, demonstrating the effective use of form validation and control structures.



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Case Study No.: 03

1. Case Study Title: Write a JavaScript program that accepts student marks and displays the scholarship category.

2. Case Study Program Code:

```
JS Practical2.js  index.html  Case study 2 java.html  JS  II  ?  ↓  ↑  ↺  □

P3 case study.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Scholarship Eligibility</title>
7  <style>
8      body{
9          font-family: Arial, sans-serif;
10         background-color: #064c96;
11         display: flex;
12         justify-content: center;
13         align-items: center;
14         height: 100vh;
15     }
16     .container{
17         background: rgb(217, 4, 4);
18         padding: 30px;
19         width: 350px;
20         text-align: center;
21         border-radius: 10px;
22         box-shadow: 0 0 10px gray;
23     }
24     h1{
25         color: navy;
26     }
27     input{
28         width: 90%;
29         padding: 10px;
30         margin: 15px 0;
31         font-size: 16px;
32     }
```

```
33     button{
34         padding: 10px 20px;
35         background: navy;
36         color: white;
37         border: none;
38         border-radius: 5px;
39         cursor: pointer;
40         font-size: 16px;
41     }
42     button:hover{
43         background: blue;
44     }
45     #result{
46         margin-top: 20px;
47         font-size: 20px;
48         font-weight: bold;
49         color: green;
```



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```
51 </style>
52 </head>
53 <body>
54 <div class="container">
55 <h1>Vithika Saoji</h1>
56 <h2>Scholarship Eligibility</h2>
57 <input type="number" id="marks" placeholder="Enter Marks">
58 <br>
59 <button onclick="checkScholarship()">Submit</button>
60 <div id="result"></div>
61 </div>
62 <script>
63 function checkScholarship()
64 {
65     let marks = document.getElementById("marks").value;
66     if(marks=="")
67     {
68         alert("Enter Marks");
69         return;
70     }
71     marks = Number(marks);
72     if(marks<0 || marks>100)
73     {
74         alert("Enter Valid Marks");
75         return;
76     }
77     let scholarship;
78     if(marks<50)
79     {
80         scholarship = "Not Eligible";
81     }
82     else if(marks>=50 && marks<=69)
83     {
84         scholarship = "Bronze Scholarship";
85     }
86     else if(marks>=70 && marks<=84)
87     {
88         scholarship = "Silver Scholarship";
```

```
89     }
90     else if(marks>=85 && marks<=94)
91     {
92         scholarship = "Gold Scholarship";
93     }
94     else
95     {
96         scholarship = "Platinum Scholarship";
97     }
98     document.getElementById("result").innerHTML =
99     "Marks: " + marks + "<br>Scholarship: " + scholarship;
100 }
101 </script>
102 </body>
103 </html>
```

3. Output:

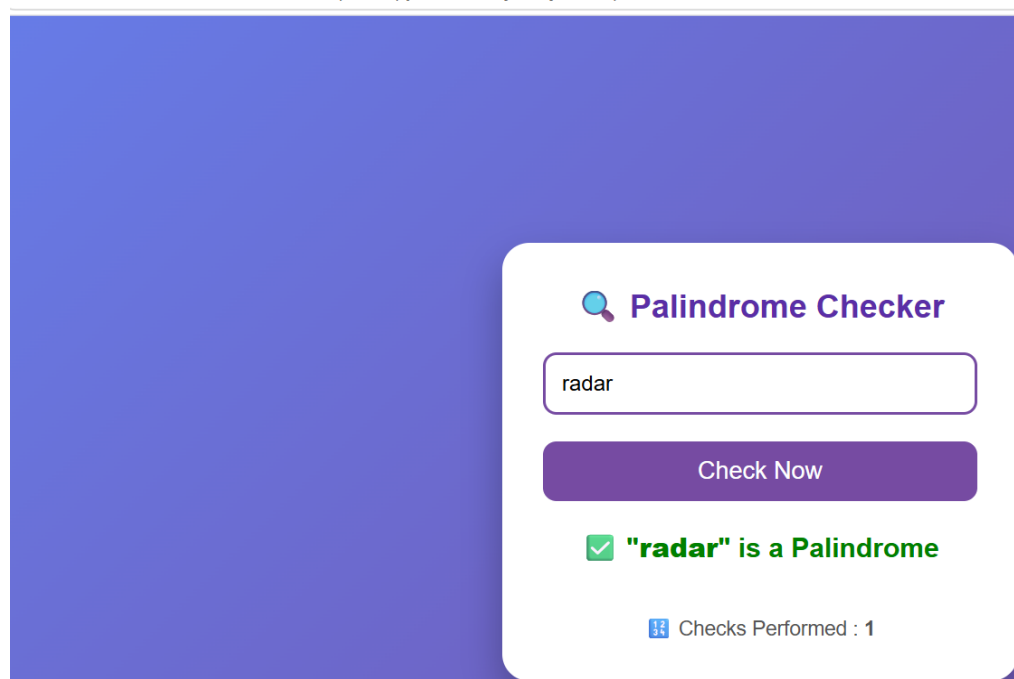
File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/P3%20case%20study.html

Experiment No.: 4

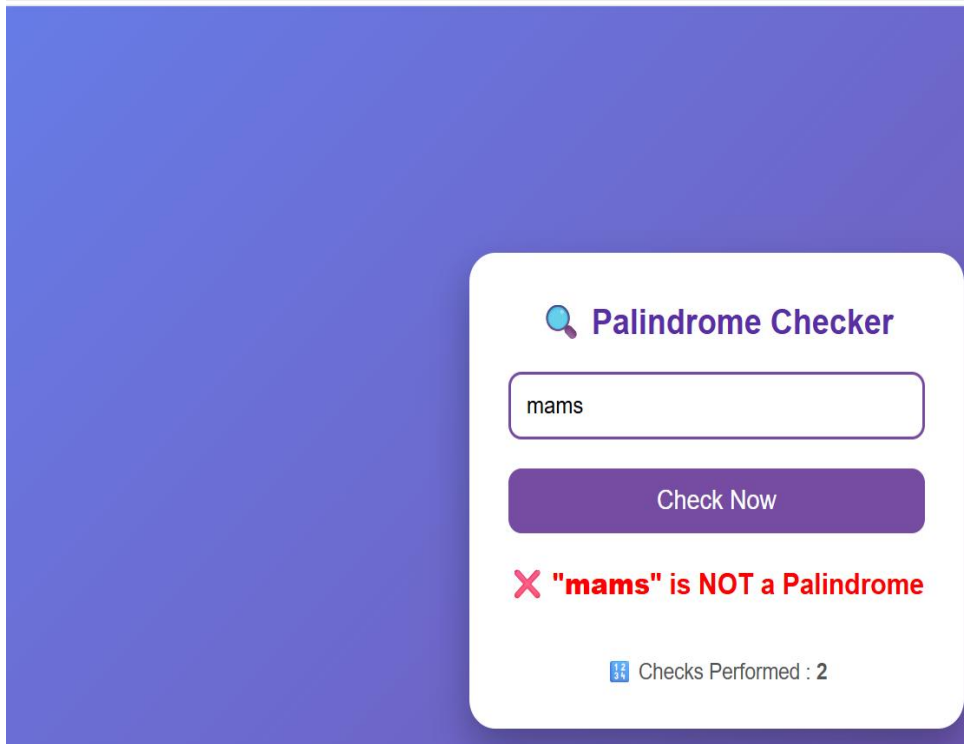
- 1. Title:** Use function types, scope, and closures; apply try catch; build a palindrome checker.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.
- 3. Theory:** This project is developed using HTML and JavaScript to create a Palindrome Checker. It demonstrates different JavaScript function types, including function declaration, function expression, and arrow functions. The program uses scope by defining global and local variables appropriately, while a closure is implemented to maintain and update the number of palindrome checks performed. Exception handling is achieved using the try...catch block, which prevents the program from crashing and displays meaningful error messages when the user enters invalid or empty input. The application accepts a word or phrase, removes unnecessary characters, checks whether it reads the same forwards and backwards, and displays the result through an interactive user interface.

4. Output :

① File C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P4%20Palindrome.html



) File C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P4%20Palindrome.html



5. Result/Conclusion: The palindrome checker was successfully developed using HTML and JavaScript. It correctly checks whether the entered text is a palindrome, handles invalid input using try-catch, demonstrates function types, scope, and closures, and displays accurate results through a simple and interactive interface. This project enhanced the understanding of core JavaScript concepts while creating a reliable, user-friendly, and error-free web application.



Case Study No.: 04

1. Case Study Title: ATM Card PIN Verification Using JavaScript Functions.

2. Case Study Program Code:

```
Release Notes: 1.131.0 P4 Palindrome.html P4 casestudy.html X
P4 casestudy.html > html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>ATM Card PIN Verification</title>
7
8 <style>
9 *{
10     margin:0;
11     padding:0;
12     box-sizing:border-box;
13     font-family:Arial, sans-serif;
14 }
15
16 body{
17     height:100vh;
18     display:flex;
19     justify-content:center;
20     align-items:center;
21     background:linear-gradient(135deg, #0f2027, #203a43, #2c5364);
22 }
23
24 .container{
25     width:360px;
26     background:#fff;
27     padding:30px;
28     border-radius:15px;
29     text-align:center;
30     box-shadow:0 0 20px rgba(0,0,0,0.3);
31 }
32
33 h2{
34     color:#2c5364;
35     margin-bottom:20px;
36 }
```

```
38 input{
39     width:100%;
40     padding:12px;
41     margin:15px 0;
42     border:2px solid #2c5364;
43     border-radius:8px;
44     font-size:18px;
45     text-align:center;
46 }
47
48 button{
49     width:100%;
50     padding:12px;
51     background:#2c5364;
52     color:white;
53     border:none;
54     border-radius:8px;
55     font-size:18px;
56     cursor:pointer;
57 }
58
59 button:hover{
60     background:#1b3d4f;
61 }
62
63 #result{
64     margin-top:20px;
65     font-size:18px;
66     font-weight:bold;
67 }
68
69 #attempts{
70     margin-top:10px;
71     color:#555;
72 }
73 </style>
74
```

```

74
75 </head>
76 <body>
77
78 <div class="container">
79
80 <h2>🏧 ATM PIN Verification</h2>
81
82 <input type="password" id="pin" maxlength="4" placeholder="Enter 4-Digit PIN">
83
84 <button onclick="verifyPIN()">Verify PIN</button>
85
86 <div id="result"></div>
87 <div id="attempts"></div>
88
89 </div>
90
91 <script>
92
93 // Global Scope
94 const correctPIN = "1234";
95
96 // Function Declaration
97 function getPIN(){
98     return document.getElementById("pin").value;
99 }
100
101 // Function Expression
102 const checkPIN = function(pin){
103     return pin === correctPIN;
104 };
105
106 // Closure
107 function loginCounter(){
108     let attempts = 0;
109

```

```

110         return function(){
111             attempts++;
112             return attempts;
113         }
114     }
115
116     const countAttempts = loginCounter();
117
118     // Arrow Function
119     const displayMessage = (message,color)=>{
120         document.getElementById("result").innerHTML = message;
121         document.getElementById("result").style.color = color;
122     };
123
124     // Main Function
125     function verifyPIN(){
126
127         try{
128
129             let pin = getPIN();    // Local Scope
130
131             if(pin===""){
132                 throw "Please enter your PIN.";
133             }
134
135             if(pin.length!=4 || isNaN(pin)){
136                 throw "PIN must contain exactly 4 digits.";
137             }
138
139             if(checkPIN(pin)){
140                 displayMessage("✅ PIN Verified Successfully!","green");
141             }
142             else{
143                 displayMessage("❌ Incorrect PIN.","red");
144             }
145

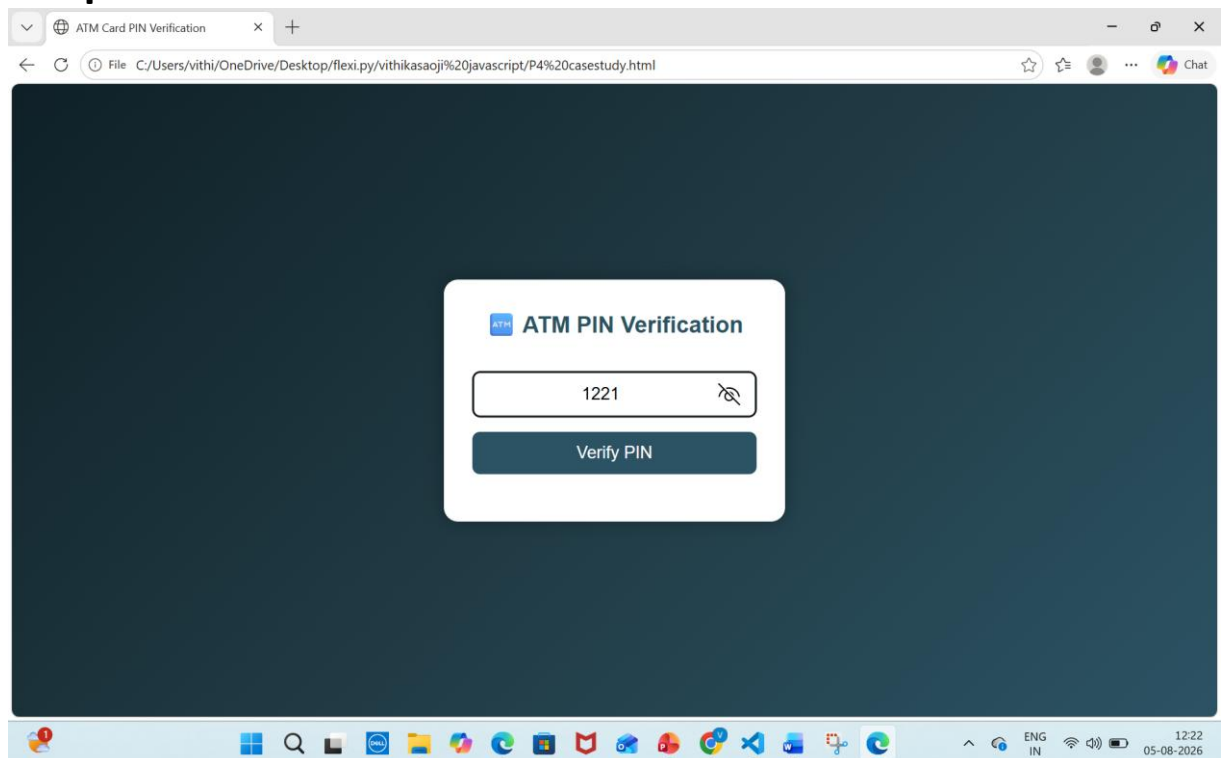
```

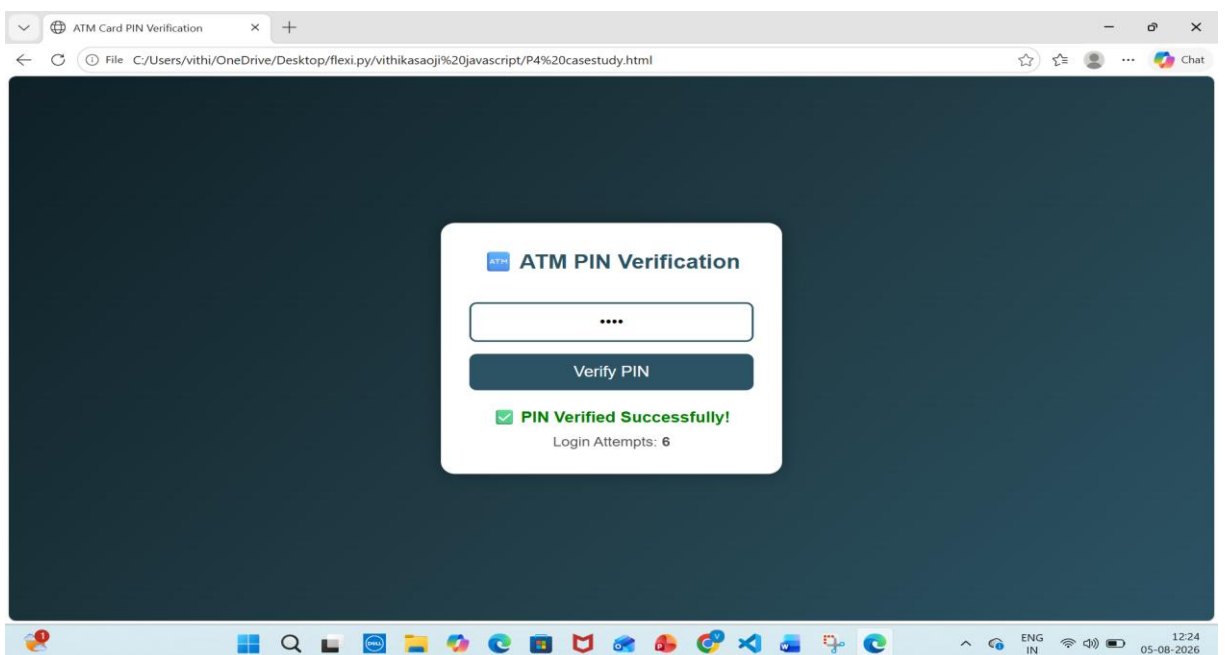
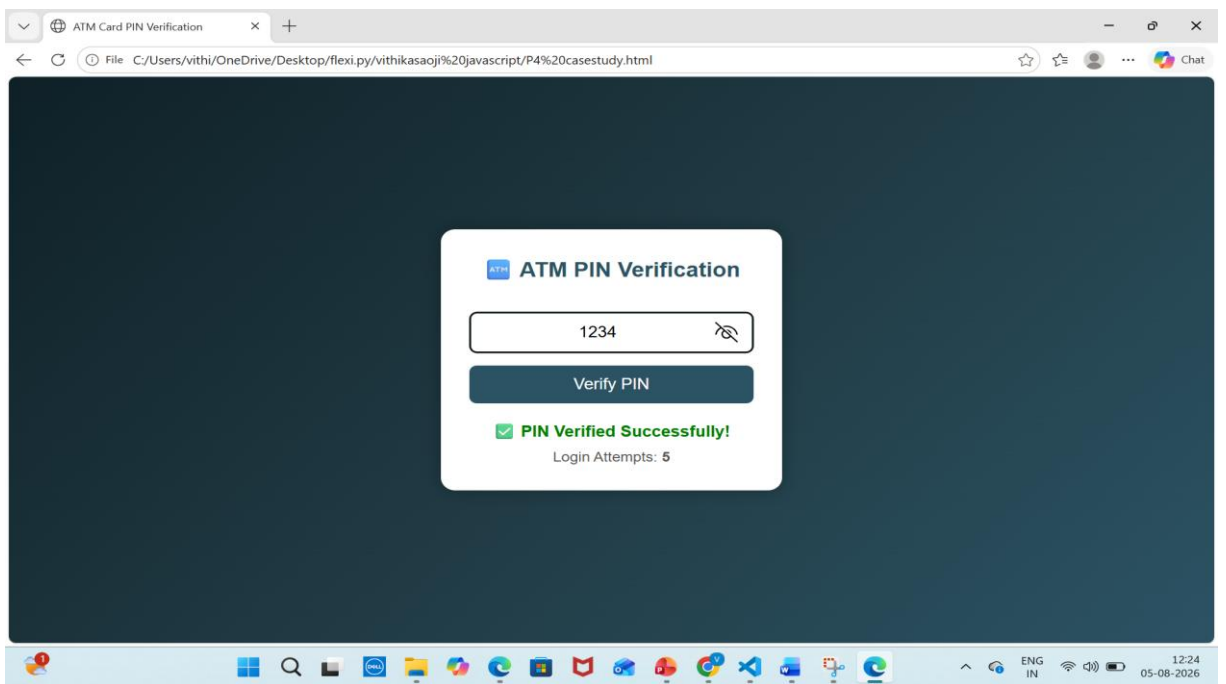
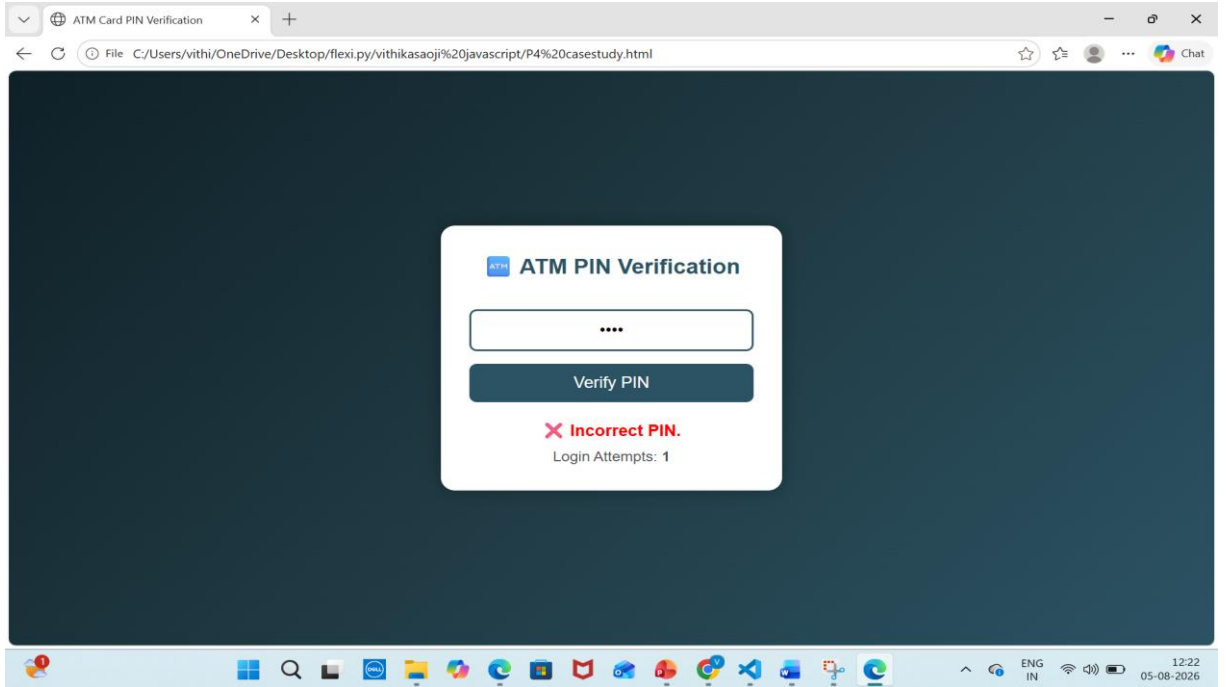
```

140     displayMessage("✅ PIN Verified Successfully.", "green");
141 }
142 else{
143     displayMessage("❌ Incorrect PIN.", "red");
144 }
145
146 document.getElementById("attempts").innerHTML =
147 "Login Attempts: <b>" + countAttempts() + "</b>";
148
149 }
150
151 catch(error){
152
153     displayMessage("⚠️ " + error, "orange");
154
155 }
156
157 finally{
158
159     console.log("PIN Verification Completed.");
160
161 }
162
163 }
164
165 </script>
166
167 </body>
168 </html>

```

3. Output:





4. Result & Conclusion: The ATM Card PIN Verification system was successfully implemented using HTML and JavaScript. It accurately verifies the entered PIN, handles invalid input using try-catch, demonstrates the use of function types, scope, and closures, and provides appropriate user feedback. This project helps in understanding core JavaScript concepts while creating a secure, reliable, and interactive PIN verification application.

Experiment No.: 5

- 1. Title:** Apply array methods and object handling; create a cart total calculator with discount logic.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.
- 3. Theory:** The Shopping Cart Total Calculator is a web application developed using HTML, CSS, and JavaScript. It demonstrates the use of arrays and objects by storing multiple products as objects inside an array. The application accepts the product name, price, and quantity from the user, calculates the total amount using the `reduce()` array method, and applies a dynamic discount based on the total purchase amount. It also displays the final payable amount through a simple, attractive, and user-friendly interface.
- 4. Output :**

le C:/Users/vithi/OneDrive/Desktop/flexi.py/vithिकासोजी%20javascript/P5.html

 **Smart Cart**
VITHIKA SAOJI
 ✨ Add items and watch the discount grow automatically ✨

 Add Item

Item Name

Price (₹)

Quantity

+ Add to Cart

 Your Cart

ITEM	PRICE	QTY	SUBTOTAL	
Laptop	₹70000.00	1	₹70000.00	✕ Remove
Headphones	₹1500.00	1	₹1500.00	✕ Remove

 Bill Summary

TOTAL AMOUNT

₹71500.00

DISCOUNT (20%)

₹14300.00

FINAL AMOUNT PAYABLE

₹57200.00

Discount Tiers • Above ₹5000: 20% | Above ₹2000: 15% | Above ₹1000: 10% | Above ₹500: 5% | Below ₹500: 0%

Mouza - Wathoda, Nagpur 440008, Maharashtra, India. | Tel: +91-712-6192370 / +91-712-6192368
 Email: info@sitnagpur.siu.edu.in | Website: www.sitnagpur.edu.in



C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P5.html

Smart Cart
VITHIKA SAOJI
 ✨ Add items and watch the discount grow automatically ✨

Add Item

Item Name:
 Price (₹):
 Quantity:
+ Add to Cart

Your Cart

ITEM	PRICE	QTY	SUBTOTAL	
Phone charger	₹100.00	1	₹100.00	<button>✕ Remove</button>
mouse	₹450.00	2	₹900.00	<button>✕ Remove</button>

Bill Summary

TOTAL AMOUNT
₹1000.00

DISCOUNT (5%)
₹50.00

FINAL AMOUNT PAYABLE
₹950.00

5. Result/Conclusion: The Shopping Cart application was successfully developed and executed. It accepts product details from the user, stores them using an array of objects, calculates the total amount using the `reduce()` method, and applies dynamic discounts based on the purchase amount. The application accurately displays the cart details, discount, and final payable amount. This project successfully demonstrates the practical implementation of array methods, object handling, user input, and discount calculation in JavaScript while providing a simple and interactive shopping cart interface.

Case Study No.: 05

1. Case Study Title: To study the functioning of Arrays and Loops in JavaScript

- Create and manipulate arrays using push(), pop(), shift(), unshift(), splice(), and slice().
- Use array methods map(), filter(), reduce(), and forEach().

Case Study: Create an array of numbers and find the maximum and minimum values.

2. Case Study Program Code:

```
P5.html x P5 case study.html x P6 email validation.html x
P5 case study.html > html > body > script > shiftElement
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Array Operations in JavaScript</title>
5
6 <style>
7     body {
8         background-color: lightblue;
9         font-family: Arial, sans-serif;
10        text-align: center;
11        padding: 40px;
12    }
13
14    .container {
15        background-color: white;
16        width: 600px;
17        margin: auto;
18        padding: 30px;
19        border-radius: 15px;
20    }
21
22    input {
23        padding: 10px;
24        width: 400px;
25        margin: 10px;
26        font-size: 16px;
27    }
28
29    button {
30        padding: 10px 18px;
31        margin: 5px;
32        border: none;
33        border-radius: 5px;
34        background-color: darkblue;
35        color: white;
36        cursor: pointer;
37    }
38
39    button:hover {
40        background-color: navy;
41    }
42
43    #output {
44        margin-top: 20px;
45        font-size: 18px;
46        text-align: left;
47        padding: 15px;
48    }
49 </style>
50 </head>
```

```

52 <body>
53
54 <div class="container">
55
56   <h2>Array Creation and Manipulation</h2>
57   <p>Enter numbers separated by commas:</p>
58
59   <input type="text" id="numbers"
60     placeholder="Example: 25, 10, 95, 5, 30, 15">
61
62   <br>
63   <button onclick="createArray()">Create Array</button>
64   <br><br>
65
66   <button onclick="pushElement()">Push</button>
67   <button onclick="popElement()">Pop</button>
68   <button onclick="shiftElement()">Shift</button>
69   <button onclick="unshiftElement()">Unshift</button>
70   <button onclick="spliceElement()">Splice</button>
71   <button onclick="sliceElement()">Slice</button>
72
73   <br><br>
74   <button onclick="findMaxMin()">Find Maximum & Minimum</button>
75   <div id="output"></div>
76
77 </div>
78
79 <script>
80
81   let numbers = [];
82
83   // Create Array
84   function createArray() {
85
86     let input = document.getElementById("numbers").value;
87     numbers = input.split(",").map(Number);
88     display("Array Created: " + numbers.join(", "));
89   }
90
91   // push() - adds element at the end
92   function pushElement() {
93     numbers.push(100);
94     display("After push(100): " + numbers.join(", "));
95   }
96
97   // pop() - removes last element
98   function popElement() {

```

```

P4 casestudy.html X P5.html X P5 case study.html
2 <html>
52 <body>
79 <script>
98   function popElement() {
99     numbers.pop();
100    display("After pop(): " + numbers.join(", "));
101  }
102  // shift() - removes first element
103  function shiftElement() {
104    numbers.shift();
105    display("After shift(): " + numbers.join(", "));
106  }
107  // unshift() - adds element at beginning
108  function unshiftElement() {
109    numbers.unshift(50);
110    display("After unshift(50): " + numbers.join(", "));
111  }
112  // splice() - removes/replaces elements
113  function spliceElement() {
114    numbers.splice(1, 1);
115    display("After splice(1, 1): " + numbers.join(", "));
116  }
117  // slice() - creates a portion of array
118  function sliceElement() {
119    let newArray = numbers.slice(1, 4);
120    display(
121      "Original Array: " + numbers.join(", ") +
122      "<br>After slice(1, 4): " + newArray.join(", ")
123    );
124  }
125  // Case Study - Find Maximum and Minimum
126  function findMaxMin() {
127    if (numbers.length === 0) {
128      display("Please create an array first.");
129      return;
130    }
131    let maximum = numbers[0];
132    let minimum = numbers[0];
133    for (let i = 1; i < numbers.length; i++) {
134      if (numbers[i] > maximum) {
135        maximum = numbers[i];
136      }
137      if (numbers[i] < minimum) {
138        minimum = numbers[i];
139      }
140    }
141    display(
142      "<b>Case Study Result</b><br><br>" +
143      "Array: " + numbers.join(", ") +
144      "<br>Maximum Value: " + maximum +

```

```

P4 casestudy.html X P5.html X P5 case study.html
P5 case study.html > html > body > script > findMaxMin
2 <html>
52 <body>
79 <script>
126   function findMaxMin() {
144     "<br>Maximum Value: " + maximum +
145     "<br>Minimum Value: " + minimum
146   };
147   }
148   // Display output
149   function display(message) {
150
151     document.getElementById("output").innerHTML = message;
152   }
153
154 </script>
155 </body>
156 </html>

```

3. Output:

Jsers/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P5%20case%20study.html

Array Creation and Manipulation

Enter numbers separated by commas:

Create Array

Push

Pop

Shift

Unshift

Splice

Slice

Find Maximum & Minimum

Array Created: 12, 23, 45, 89, 97

sers/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P5%20case%20study.html

Array Creation and Manipulation

Enter numbers separated by commas:

Create Array

Push

Pop

Shift

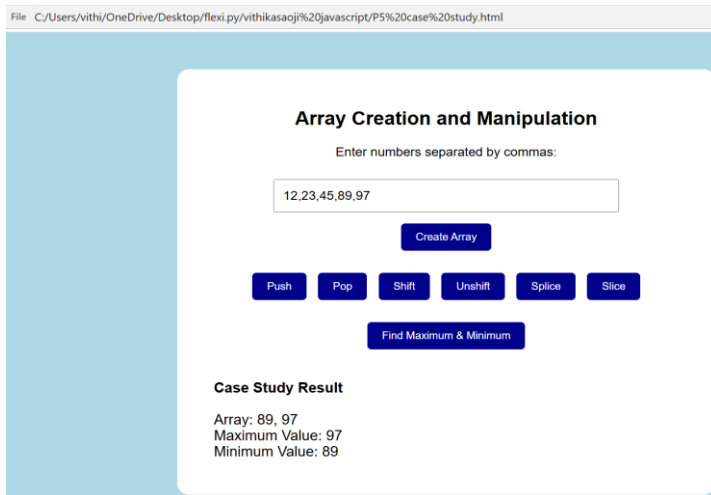
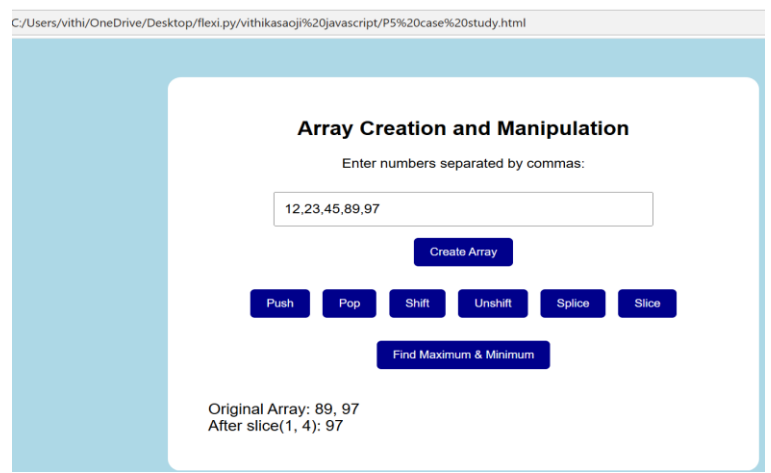
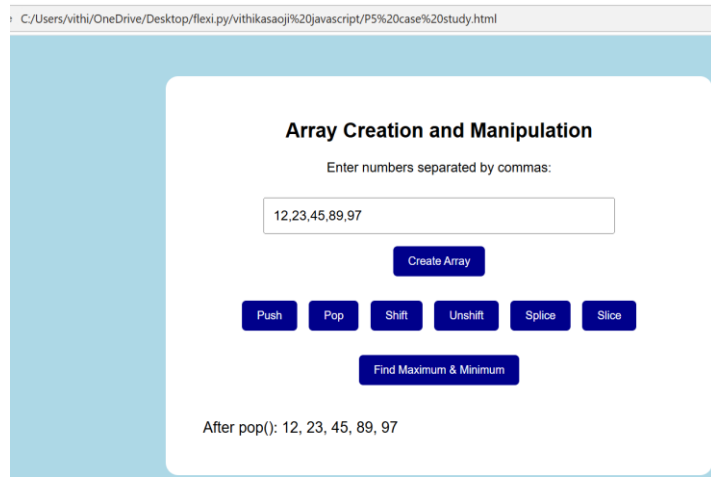
Unshift

Splice

Slice

Find Maximum & Minimum

After push(100): 12, 23, 45, 89, 97, 100



4. Result/Conclusion:

Thus, the functioning of JavaScript arrays and loops was studied and implemented successfully. Various array methods such as push(), pop(), shift(), unshift(), splice(), slice(), map(), filter(), reduce(), and forEach() were used to manipulate and process array elements. The maximum and minimum values were also successfully found from an array of numbers.

Experiment No.: 6

- 1. Title:** use string functions and regex for email validation, data extraction, and text analysis.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.
- 3. Theory:** String functions in JavaScript are used to manipulate and analyze text. In this practical, functions like `toUpperCase()`, `toLowerCase()`, `includes()`, `match()`, `split()`, and `length` are used for text analysis.

Regular Expressions (Regex): are patterns used to search, extract, and validate text. In this practical, Regex is used to extract email IDs from a paragraph and check whether they follow a valid email format.

The program also checks the presence of "Welcome", counts words and the occurrence of "email", and displays valid and invalid email IDs.

4. Output :

File C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaaji%20javascript/P6%20email%20validation.html

Email Validation & Text Analysis

Enter a paragraph to extract and validate email IDs

Welcome to our college project portal! Hello, I am Ananya, a college student working on a web development project. I usually communicate with my classmates through email. My friend Rahul uses rahul.sharma@gmail.com, while Priya can be contacted at priya123@yahoo.com. Aman gave me his email as aman@company, which is invalid because it has no proper domain extension. Neha accidentally wrote neha@.com, and Rohan entered rohan@gmail.com, both of which are invalid email formats. I also received valid email IDs such as karan.dev@outlook.com and sneha2026@proton.me. However, vikas@domain..com and meena@gmail.com are invalid. Welcome to our platform, where we share assignments, project files, and important college updates through email.

Analyze Text

 **Extracted Emails**

rahul.sharma@gmail.com, priya123@yahoo.com, aman@company, neha@.com, karan.dev@outlook.com, sneha2026@proton.me, vikas@domain..com

 **Email Validation**

false

 **Valid Emails**

rahul.sharma@gmail.com → Valid

priya123@yahoo.com → Valid

karan.dev@outlook.com → Valid

sneha2026@proton.me → Valid

vikas@domain..com → Valid



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P6%20email%20validation.html

Invalid Emails

aman@company → Invalid

neha@.com → Invalid

Uppercase

WELCOME TO OUR COLLEGE PROJECT PORTAL! HELLO, I AM ANANYA, A COLLEGE STUDENT WORKING ON A WEB DEVELOPMENT PROJECT. I USUALLY COMMUNICATE WITH MY CLASSMATES THROUGH EMAIL. MY FRIEND RAHUL USES RAHUL.SHARMA@GMAIL.COM, WHILE PRIYA CAN BE CONTACTED AT PRIYA123@YAHOO.COM. AMAN GAVE ME HIS EMAIL AS AMAN@COMPANY, WHICH IS INVALID BECAUSE IT HAS NO PROPER DOMAIN EXTENSION. NEHA ACCIDENTALLY WROTE NEHA@.COM, AND ROHAN ENTERED ROHAN.GMAIL.COM, BOTH OF WHICH ARE INVALID EMAIL FORMATS. I ALSO RECEIVED VALID EMAIL IDS SUCH AS KARAN.DEV@OUTLOOK.COM AND SNEHA2026@PROTON.ME. HOWEVER, VIKAS@DOMAIN.COM AND MEENA#GMAIL.COM ARE INVALID. WELCOME TO OUR PLATFORM, WHERE WE SHARE ASSIGNMENTS, PROJECT FILES, AND IMPORTANT COLLEGE UPDATES THROUGH EMAIL.

Lowercase

welcome to our college project portal! hello, i am ananya, a college student working on a web development project. i usually communicate with my classmates through email. my friend rahul uses rahul.sharma@gmail.com, while priya can be contacted at priya123@yahoo.com. aman gave me his email as aman@company, which is invalid because it has no proper domain extension. neha accidentally wrote neha@.com, and rohan entered rohan.gmail.com, both of which are invalid email formats. i also received valid email ids such as karan.dev@outlook.com and sneha2026@proton.me. however, vikas@domain.com and meena#gmail.com are invalid. welcome to our platform, where we share assignments, project files, and important college updates through email.

Text Analysis

738

Text Length

105

No. of Words

5

"email" Occurrence

Contains "Welcome"

true

5. Result/Conclusion: The program was successfully executed. It extracts email IDs from the given paragraph, validates them using Regex, and displays valid and invalid emails separately. It also performs uppercase/lowercase conversion, checks for "Welcome", calculates text length, counts words, and finds the occurrence of "email". Thus, string functions and regular expressions were successfully used for email validation, data extraction, and text analysis.



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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 06

1. Case Study Title: To perform basic string operations ; to reverse a string and count number of vowels in a paragraph.

2. Case Study Program Code:

<> Pract 6 case study.html > html > body

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>String Operations</title>
5 <style>
6   body {
7     background-color: rgb(171, 238, 243);
8     font-family: Arial;
9   }
10  input {
11    padding: 8px;
12    margin: 5px;
13    width: 350px;
14  }
15  textarea {
16    padding: 10px;
17    margin: 5px;
18    width: 500px;
19    height: 80px;
20  }
21  button {
22    padding: 10px 20px;
23    margin: 10px 5px;
24  }
25  #output {
26    margin-top: 20px;
27    font-size: 18px;
28  }
29 </style>
30 </head>
31 <body>
```

```
30 </head>
31 <body>
32 <h2>Vithika's JavaScript String Operations</h2>
33 <label>Enter String:</label><br>
34 <textarea id="str" placeholder="Enter your string or paragraph">
35 <br>
36 <label>Substring Start:</label>
37 <input type="number" id="start" placeholder="Enter start index">
38 <br>
39 <label>Substring End:</label>
40 <input type="number" id="end" placeholder="Enter end index">
41 <br>
42 <label>IndexOf - Enter text to search:</label>
43 <input type="text" id="search" placeholder="Enter character to search">
44 <br>
45 <label>Split - Enter separator:</label>
46 <input type="text" id="separator" placeholder="Example: space">
47 <br>
48 <label>Replace - Text to replace:</label>
49 <input type="text" id="oldText" placeholder="Enter text to replace">
50 <br>
51 <label>Replace - New text:</label>
52 <input type="text" id="newText" placeholder="Enter new text">
53 <br>
54 <button onclick="performOperations()">Perform Operations</button>
55 <div id="output"></div>
56 <script src="shreya_case6.js"></script>
57 </body>
58 </html>
```

3.Output:

Vithika's JavaScript String Operations

Enter String:

```
Technology is changing the way people live, work, and communicate.  
The internet provides easy access to information from different  
parts of the world.  
Learning new technologies can improve problem-solving and  
creativity.
```

Substring Start:

Substring End:

IndexOf - Enter text to search:

Split - Enter separator:

Replace - Text to replace:

Replace - New text:

Perform Operations

4.Result/Conclusion:

The JavaScript string operations such as substring, indexOf, split, and replace were successfully performed on the given string. The practical helped in understanding how JavaScript string methods can be used to extract, search, split, and replace text efficiently.

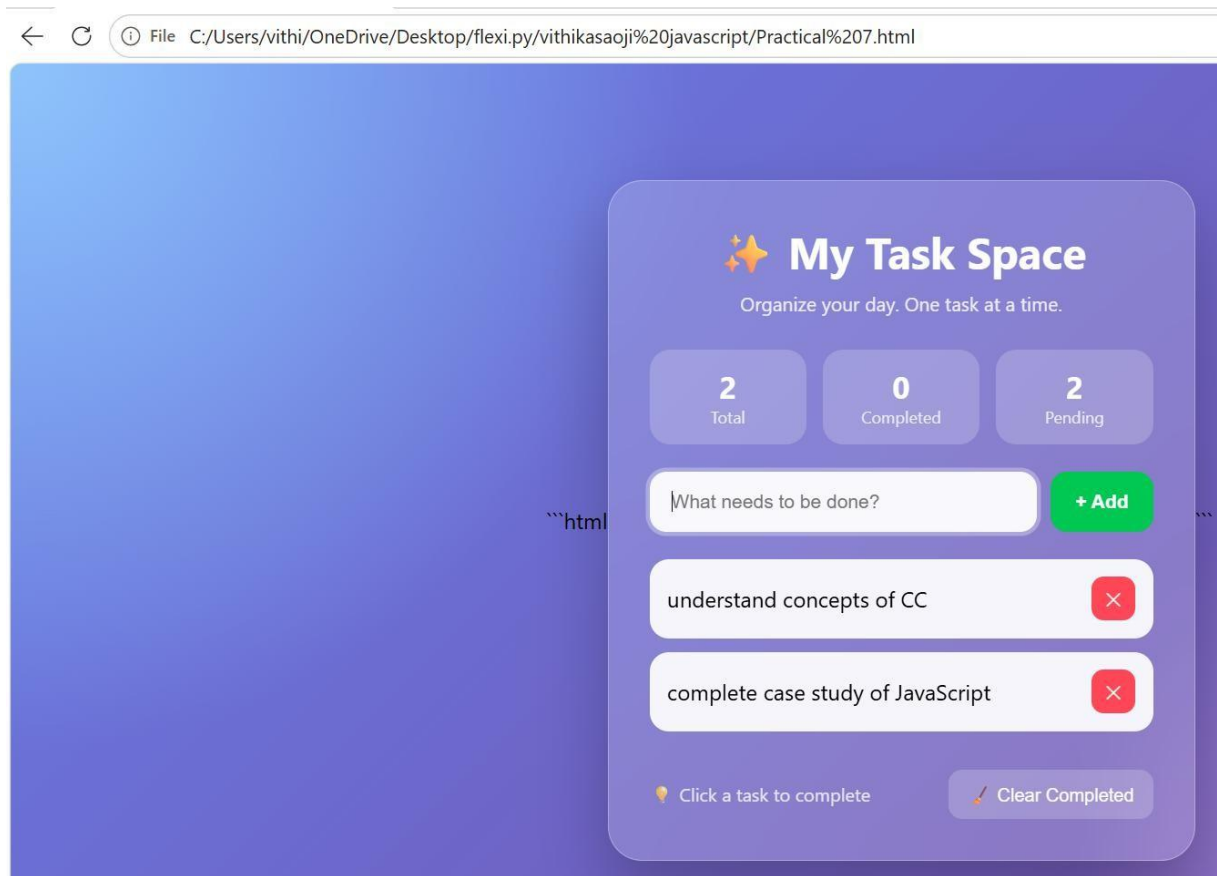
Practical No.: 7

1. Title: Perform DOM traversal and updates; develop a to do list app with add, edit, and delete features.

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: The Document Object Model (DOM) represents an HTML webpage as a tree of elements that can be accessed and modified using JavaScript. DOM traversal is the process of navigating through HTML elements using properties and methods such as `parentElement`, `children`, `nextElementSibling`, `querySelector()`, and `querySelectorAll()`. DOM updates allow JavaScript to dynamically change the content, style, and structure of a webpage without reloading it. In a To-Do List application, JavaScript is used to access the input field and task list, create new task elements using `createElement()`, and add them to the DOM using `appendChild()`. Users can add new tasks, edit existing tasks, and delete unwanted tasks dynamically. Event listeners such as `click` are used to detect user actions and perform the required operations. The `textContent` property is used to display or modify task text, while methods like `remove()` can delete elements from the DOM. Thus, the To-Do List demonstrates practical use of DOM traversal, DOM manipulation, event handling, and dynamic webpage updates using JavaScript.

Output :



- 4. Result/Conclusion:** The To-Do List application was successfully developed using HTML, CSS, and JavaScript. DOM traversal and manipulation were performed to dynamically add, edit, and delete tasks. The application demonstrates how JavaScript can interact with HTML elements and update webpage content dynamically without reloading the page.

Case Study No.: 07

1. Case Study Title: To Demonstrate the Events Handling in JavaScript

- a) Access and manipulate form elements using JavaScript.
- b) Validate user input (e.g., email, empty fields).
- c) Use various form events (submit, change, focus).

Create the following HTML form and use required JavaScript and events in a given form.

2. Case Study Program Code:

```
<> Case study 7.html > ...
1  <html>
2  <!DOCTYPE html>
3  <html>
4  <head>
5    <title>JavaScript Events Handling</title>
6
7    <style>
8      body {
9        font-family: Arial;
10       background: #eef3f8;
11       display: flex;
12       justify-content: center;
13       padding: 40px;
14     }
15
16     .form-box {
17       width: 400px;
18       background: white;
19       padding: 25px;
20       border-radius: 10px;
21       box-shadow: 0 4px 15px #bbb;
22     }
23
24     h2 {
25       text-align: center;
26       color: #245a8d;
27     }
28
29     label {
30       display: block;
31       margin-top: 15px;
32       font-weight: bold;
33     }
34
35     input, select, textarea {
36       width: 100%;
37       padding: 10px;
38       margin-top: 5px;
39       box-sizing: border-box;
40       border: 1px solid #aaa;
41       border-radius: 5px;
42     }
43
44     textarea {
45       resize: none;
46       height: 70px;
47     }
48
49     button {
50       width: 100%;
```

```

50       width: 100%;
51       padding: 11px;
52       margin-top: 20px;
53       background: #245a8d;
54       color: white;
55       border: none;
56       border-radius: 5px;
57       cursor: pointer;
58     }
59
60     .error {
61       color: red;
62       font-size: 13px;
63     }
64
65     .success {
66       color: green;
67       margin-top: 15px;
68       font-weight: bold;
69     }
70   </style>
71 </head>
72
73 <body>
74
75   <div class="form-box">
76
77     <h2>Student Registration Form</h2>
78
79     <form id="myForm">
80
81       <label>Name</label>
82       <input type="text" id="name" placeholder="Enter your name">
83       <div id="nameError" class="error"></div>
84
85       <label>Email</label>
86       <input type="text" id="email" placeholder="Enter your email">
87       <div id="emailError" class="error"></div>
88
89       <label>Age</label>
90       <input type="number" id="age" placeholder="Enter your age">
91       <div id="ageError" class="error"></div>
92
93       <label>Course</label>
94       <select id="course">
95         <option value="">Select Course</option>
```

```

96         <option value="CSE">CSE</option>
97         <option value="IT">IT</option>
98         <option value="ECE">ECE</option>
99     </select>
100
101     <label>Address</label>
102     <textarea id="address" placeholder="Enter your address"></textarea>
103
104     <button type="submit">Submit</button>
105
106     <div id="result" class="success"></div>
107
108 </form>
109
110 </div>
111
112 <script>
113     // Accessing form elements
114     let name = document.getElementById("name");
115     let email = document.getElementById("email");
116     let age = document.getElementById("age");
117     let course = document.getElementById("course");
118     let address = document.getElementById("address");
119     let form = document.getElementById("myForm");
120
121     // Focus Event
122     name.addEventListener("focus", function() {
123         name.style.backgroundColor = "#eef7ff";
124     });
125
126     // Input Event - Name validation
127     name.addEventListener("input", function() {
128         if (name.value.trim() == "") {
129             document.getElementById("nameError").innerText =
130                 "Invalid: Name is required";
131         } else {
132             document.getElementById("nameError").innerText = "";
133         }
134     });
135
136     // Input Event - Email validation
137     email.addEventListener("input", function() {
138         let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
139         if (!pattern.test(email.value)) {
140             document.getElementById("emailError").innerText =
141                 "Invalid: Enter a valid email";
142         } else {
143             document.getElementById("emailError").innerText = "";
144         }
145     });
146
147     // Input Event - Age validation
148     age.addEventListener("input", function() {
149         if (age.value == "" || age.value < 18 || age.value > 60) {
150             document.getElementById("ageError").innerText =
151                 "Invalid: Age must be between 18 and 60";
152         } else {
153             document.getElementById("ageError").innerText = "";
154         }
155     });
156
157     course.addEventListener("change", function() {
158         document.getElementById("result").innerText =
159             "Selected Course: " + course.value;
160     });
161
162     form.addEventListener("submit", function(event) {
163         event.preventDefault();
164         let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
165         if (
166             name.value.trim() == "" ||
167             !emailPattern.test(email.value) ||
168             age.value < 18 ||
169             age.value > 60 ||
170             course.value == "" ||
171             address.value.trim() == ""
172         ) {
173             alert("Please enter valid details.");
174             return;
175         }
176
177         document.getElementById("result").innerText =
178             "Form submitted successfully! Welcome " + name.value + ".";
179     });
180
181 </script>
182
183 </body>
184 </html>
185
186

```

```

140 // Input Event - Email validation
141 email.addEventListener("input", function() {
142
143     let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
144
145     if (!pattern.test(email.value)) {
146         document.getElementById("emailError").innerText =
147             "Invalid: Enter a valid email";
148     } else {
149         document.getElementById("emailError").innerText = "";
150     }
151
152 });
153
154 // Input Event - Age validation
155 age.addEventListener("input", function() {
156
157     if (age.value == "" || age.value < 18 || age.value > 60) {
158         document.getElementById("ageError").innerText =
159             "Invalid: Age must be between 18 and 60";
160     } else {
161         document.getElementById("ageError").innerText = "";
162     }
163
164 });
165
166 course.addEventListener("change", function() {
167
168     document.getElementById("result").innerText =
169         "Selected Course: " + course.value;
170
171 });
172
173 form.addEventListener("submit", function(event) {
174
175     event.preventDefault();
176     let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
177     if (
178         name.value.trim() == "" ||
179         !emailPattern.test(email.value) ||
180         age.value < 18 ||
181         age.value > 60 ||
182         course.value == "" ||
183         address.value.trim() == ""
184     ) {
185         alert("Please enter valid details.");
186         return;
187     }
188
189     document.getElementById("result").innerText =
190         "Form submitted successfully! Welcome " + name.value + ".";
191
192 });
193
194 </script>
195
196 </body>
197 </html>
198
199

```

```

187     document.getElementById("result").innerText =
188         "Form submitted successfully! Welcome " + name.value + ".";
189
190 });
191
192 </script>
193
194 </body>
195 </html>
196
197

```

3. Output:

~/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/Case%20study%207.html

Student Registration Form

Name

Riya Sharma

Email

Riya.sharmagmail.com

Invalid: Enter a valid email

Age

10

Invalid: Age must be between 18 and 60

Course

IT

Address

25 MG Road, Nagpur

Submit

Student Registration Form

Name

Riya Sharma

Email

Riya.sharma@gmail.com

Age

20

Course

IT

Address

25 MG Road, Nagpur

Submit

Form submitted successfully! Welcome Riya Sharma .

4. Result / Conclusion:

The HTML form was successfully created using JavaScript event handling. The form elements were accessed and validated using JavaScript, and focus, change, and submit events were successfully implemented.

Practical No.: 8

- 1. Title:** Access and validate form fields using events; design a gym admission form with live input checks.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.
- 3. Theory:** In this practical, a Gym Admission Form is designed using HTML, CSS, and JavaScript. JavaScript is used to access and validate form fields using events. The input values are accessed using the .value property, such as name.value, mobile.value, email.value, and age.value. Live validation is performed using the input event, so the user gets immediate feedback while entering data. The mobile number is checked to ensure that it contains exactly 10 digits. The email field is validated using a proper email pattern. The age is restricted between 15 and 50 years. A membership plan is selected using a dropdown containing Monthly, Quarterly, and Yearly options. Invalid inputs are highlighted and an invalid message is displayed. The form uses the submit event to validate all fields before submission. After successful submission, an acknowledgment message is displayed with the member's entered details.
- 4. Output:**

File C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/Practical%208.html

html

**Vithika's Gym Admission Form**

Name

Mobile Number

Email

Age

Membership Plan

Submit Admission

'''html

 **Vithika's Gym Admission Form**

Name

Mobile Number

Invalid: Mobile number must be 10 digits

Email

Invalid: Enter a valid email

Age

Invalid: Age must be between 15 and 50

Membership Plan

Submit Admission

 **Admission Successful!**

Welcome, **Riya Sharma!**

Mobile: **9479473488**

Email: **riyasharma@gmail.com**

Age: **25**

Membership: **Monthly**

Thank you for joining our gym! 🍌

5. Result/Conclusion: The Gym Admission Form was successfully created with live validation using JavaScript events. It checks the name, 10-digit mobile number, email, age (15–50), and membership plan. Invalid inputs show error messages, and after valid submission, an acknowledgment message is displayed.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.:08

1. Case Study Title: Create an Online Job Application Form Using Javascript.

2. Case Study Program Code:

```
Case study8.html > html > head > style > input > select > textarea
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Online Job Application Form</title>
5
6 <style>
7   body {
8     font-family: Arial, sans-serif;
9     background: linear-gradient(135deg, #dbefe7, #e0e7ff);
10    margin: 0;
11    padding: 25px;
12  }
13
14  .container {
15    width: 650px;
16    margin: auto;
17    background: #fff;
18    padding: 25px;
19    border-radius: 15px;
20    box-shadow: 0 5px 20px rgba(0,0,0,0.15);
21  }
22
23  h2 {
24    text-align: center;
25    color: #3730a3;
26  }
27
28  h3 {
29    color: #4f46e5;
30    border-bottom: 2px solid #ddd;
31    padding-bottom: 5px;
32    margin-top: 25px;
33  }
34
35  label {
36    display: block;
37    margin-top: 12px;
38    font-weight: bold;
39  }
40
41  input, select, textarea {
42    width: 100%;
43    padding: 10px;
44    margin-top: 5px;
45    border: 1px solid #ccc;
46    border-radius: 6px;
47    box-sizing: border-box;
48  }
49
50  textarea {
51    height: 70px;
52    resize: none;
53  }
54
55  .radio {
56    width: auto;
57    margin-right: 5px;
58  }
59
60  .message {
```

```

94 </style>
95 </head>
96
97 <body>
98
99 <div class="container">
100
101   <h2> Online Job Application Form</h2>
102
103   <form id="jobForm">
104
105     <h3>Personal Information</h3>
106
107     <!-- NAME -->
108     <label>Full Name:</label>
109     <input type="text" id="name" placeholder="Enter full name">
110     <div id="nameMessage" class="message"></div>
111
112     <!-- DOB -->
113     <label>Date of Birth:</label>
114     <input type="date" id="dob">
115     <div id="dobMessage" class="message"></div>
116
117     <!-- GENDER -->
118     <label>Gender:</label>
119     <input type="radio" class="radio" name="gender" value="Male"> Male
120     <input type="radio" class="radio" name="gender" value="Female"> Female
121     <input type="radio" class="radio" name="gender" value="Other"> Other
122     <div id="genderMessage" class="message"></div>
123
124     <!-- EMAIL -->
125     <label>Email Address:</label>
126     <input type="text" id="email" placeholder="Enter email">
127     <div id="emailMessage" class="message"></div>
128
129     <!-- MOBILE -->
130     <label>Mobile Number:</label>
131     <input type="text" id="mobile"
132           placeholder="Enter 10-digit mobile number"
133           maxlength="10">
134     <div id="mobileMessage" class="message"></div>
135
136     <!-- ADDRESS -->
137     <label>Address:</label>
138     <textarea id="address" placeholder="Enter address"></textarea>
139     <div id="addressMessage" class="message"></div>
140
141     <!-- CITY -->
142     <label>City:</label>
143     <input type="text" id="city" placeholder="Enter city">
144     <div id="cityMessage" class="message"></div>
145
146     <!-- STATE -->
147     <label>State:</label>
148     <input type="text" id="state" placeholder="Enter state">
149     <div id="stateMessage" class="message"></div>
150
```

```

151 <!-- PIN -->
152 <label>PIN Code:</label>
153 <input type="text" id="pin"
154       placeholder="Enter 6-digit PIN"
155       maxlength="6">
156 <div id="pinMessage" class="message"></div>
157
158 <!-- QUALIFICATION -->
159 <label>Educational Qualification:</label>
160 <select id="qualification">
161   <option value="">Select qualification</option>
162   <option>B.Tech</option>
163   <option>B.E.</option>
164   <option>BCA</option>
165   <option>MCA</option>
166   <option>M.Tech</option>
167   <option>Other</option>
168 </select>
169 <div id="qualificationMessage" class="message"></div>
170
171 <!-- PERCENTAGE -->
172 <label>Percentage:</label>
173 <input type="number" id="percentage"
174       placeholder="Enter percentage">
175 <div id="percentageMessage" class="message"></div>
176
177 <!-- PASSING YEAR -->
178 <label>Passing Year:</label>
179 <input type="number" id="passingYear"
180       placeholder="Enter passing year">
181 <div id="yearMessage" class="message"></div>
182
183
184 <h3>Professional Information</h3>
185
186 <!-- POSITION -->
187 <label>Applying for Job Position:</label>
188 <input type="text" id="position"
189       placeholder="Example: Software Developer">
190 <div id="positionMessage" class="message"></div>
191
192 <!-- EXPERIENCE -->
193 <label>Total Work Experience (Years):</label>
194 <input type="number" id="experience"
195       placeholder="Enter experience">
196 <div id="experienceMessage" class="message"></div>
197
198 <!-- ORGANIZATION -->
199 <label>Current / Previous Organization:</label>
200 <input type="text" id="organization"
201       placeholder="Enter organization">
202 <div id="organizationMessage" class="message"></div>
203
204 <!-- DESIGNATION -->
205 <label>Current Designation:</label>
206 <input type="text" id="designation"

```

```

542
543     form.reset();
544
545   } else {
546
547     document.getElementById("result").className = "error";
548
549     document.getElementById("result").innerHTML =
550       "❌ Please correct the invalid details before submitting.";
551
552   }
553
554 });
555
556 </script>
557
558 </body>
559 </html>

```

```

488 form.addEventListener("submit", function(event) {
489
490
491     event.preventDefault();
492
493     let name = document.getElementById("name").value.trim();
494     let dob = document.getElementById("dob").value;
495     let gender = document.querySelector('input[name="gender"]:checked');
496     let email = document.getElementById("email").value;
497     let mobile = document.getElementById("mobile").value;
498     let address = document.getElementById("address").value.trim();
499     let city = document.getElementById("city").value.trim();
500     let state = document.getElementById("state").value.trim();
501     let pin = document.getElementById("pin").value;
502     let qualification = document.getElementById("qualification").value;
503     let percentage = Number(document.getElementById("percentage").value);
504     let passingYear = Number(document.getElementById("passingYear").value);
505
506     let position = document.getElementById("position").value.trim();
507     let experience = Number(document.getElementById("experience").value);
508     let organization = document.getElementById("organization").value.trim();
509     let designation = document.getElementById("designation").value.trim();
510     let salary = Number(document.getElementById("salary").value);
511     let skills = document.getElementById("skills").value.trim();
512
513     let emailPattern = /^[^s@]+@[^s@]+\.[^s@]+$/;
514     let currentYear = new Date().getFullYear();
515
516     if (
517         name !== "" &&
518         dob !== "" &&
519         gender &&
520         emailPattern.test(email) &&
521         mobile.length === 10 &&
522         address !== "" &&
523         city !== "" &&
524         state !== "" &&
525         pin.length === 6 &&
526         qualification !== "" &&
527         percentage >= 0 &&
528         percentage <= 100 &&
529         passingYear >= 1950 &&
530         passingYear <= currentYear &&
531         position !== "" &&
532         experience >= 0 &&
533         organization !== "" &&
534         designation !== "" &&
535         salary > 0 &&
536         skills !== ""
537     ) {
538         document.getElementById("result").className = "success";
539
540         document.getElementById("result").innerHTML =
541             "✅ Application Submitted Successfully!";

```

3.Output:

 **Online Job Application Form**

Personal Information

Full Name:

✓ Valid name

Date of Birth:

Date of birth is required

Gender:
☐ Male ☒ Female ☐ Other
✓ Gender selected

Email Address:

Invalid email address

Mobile Number:

Invalid mobile number - enter 10 digits

Address:

✓ Valid address

City:

✓ Valid city

State:

✓ Valid state

Expected Salary:

✓ Valid expected salary

Technical Skills:

✓ Skills entered

[Submit Application](#)

✓ Application Submitted Successfully!

✓ Valid city

State:

✓ Valid state

PIN Code:

Invalid PIN - enter 6 digits

Educational Qualification:

✓ Qualification selected

Percentage:

✓ Valid percentage

Passing Year:

✓ Valid passing year

Professional Information

Applying for Job Position:

✓ Valid job position

Total Work Experience (Years):

✓ Valid experience

Current / Previous Organization:

✓ Valid organization

Current Designation:

✓ Valid designation

Expected Salary:

✓ Valid expected salary

Technical Skills:

✓ Skills entered

[Submit Application](#)

4. Result / Conclusion:

The Online Job Application Form was successfully created using HTML, CSS, and JavaScript. JavaScript was used to validate the applicant's personal, contact, and educational details before submission. The form checks whether the entered information is valid and displays appropriate messages for incorrect or missing details. Thus, the case study demonstrates how JavaScript can be used for form validation, error handling, and creating an interactive web-based application form.

